

An open-source, high resolution, automated fluorescence microscope

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Abstract

Fluorescence microscopy is a fundamental tool in the life sciences, but the availability of sophisticated equipment required to yield high-quality, quantitative data is a major bottleneck in data production in many laboratories worldwide. This problem has long been recognized and the abundance of low-cost electronics and the simplification of fabrication through 3D-printing have led to the emergence of open-source scientific hardware as a research field. Cost effective fluorescence microscopes can be assembled from cheaply mass-produced components, but lag behind commercial solutions in image quality. On the other hand, blueprints of sophisticated microscopes such as light-sheet or super-resolution systems, custom-assembled from high quality parts, are available, but require a high level of expertise from the user. Here we combine the UC2 microscopy toolbox with high-quality components and integrated electronics and software to assemble an automated high-resolution fluorescence microscope. Using this microscope, we demonstrate high resolution fluorescence imaging for fixed and live samples. When operated inside an incubator, long-term live-cell imaging over several days was possible. Our microscope reaches single molecule sensitivity, and we performed single particle tracking and SMLM super-resolution microscopy experiments in cells. Our setup costs a fraction of its commercially available counterparts but still provides a maximum of capabilities and image quality. We thus provide a proof of concept that high quality scientific data can be generated by lay users with a low-budget system and open-source software. Our system can be used for routine imaging in laboratories that do not have the means to acquire commercial systems and through its affordability can serve as teaching material to students.

eLife assessment

This **important** study provides **compelling** evidence that the low-cost and open-hardware UC2 microscopy framework can be expanded to enable single-molecule localization microscopy. The authors managed to fit the instrumentation and control thereof in a unit that can be placed in a small stage-top-incubator. Together with providing adapted software for data acquisition and data analysis, the UC.STORM setup can rival the capabilities of comparable commercial instruments at a fraction of the costs.

Introduction

Fluorescence microscopy is, due to its excellent contrast, specificity and compatibility with live cell imaging, an essential tool in the life sciences. Digital imaging, quantitative analysis and ever higher throughput and resolution in new techniques make fluorescence microscopy more important than ever. Nevertheless, while the development of high-end instrumentation pushes forward our progress in understanding cellular processes, spatial and temporal resolution are not the only challenges impeding progress in scientific discovery. This is because necessarily, the access to cutting-edge instrumentation is reserved to a subset of scientists as a result of its cost and technical complexity, effectively limiting data production to laboratories with access to substantial resources. At the same time, cost of electronic or optical components, powerful processors and high-quality cameras are at an all-time low. Together with simple prototyping through 3D-printing, this has led to the emergence of open-source scientific hardware as a research field (Wenzel, 2023 [↗](#)). The open-source microscopy community has experienced the rapid development of different approaches aiming to make microscopy more available (Almada et al., 2019 [↗](#); Alsamsam et al., 2022 [↗](#); Ambrose et al., 2020 [↗](#); Auer et al., 2018 [↗](#); Danial et al., 2022 [↗](#); Diederich et al., 2019a [↗](#); Hohlbein et al., 2022 [↗](#); Sharkey et al., 2016 [↗](#); Voigt et al., 2019 [↗](#); Wenzel, 2023 [↗](#)).

The first iterations of open-source microscopes came from microscopy laboratories that tried to provide similar quality data as commercial systems while being assembled from components for lower overall cost, however still requiring significant expertise. Solutions now exist for single molecule localization microscopy (SMLM) (Alsamsam et al., 2022 [↗](#); Auer et al., 2018 [↗](#); Holm et al., 2014 [↗](#); Kwakwa et al., 2016 [↗](#); Martens et al., 2019 [↗](#)), they include modular platforms (H. Li et al., 2020 [↗](#)), light sources (Berry et al., 2021 [↗](#); Schröder et al., 2020 [↗](#)) and even sophisticated special solutions such as light sheet microscopy (Voigt et al., 2019 [↗](#)), spinning disc microscopes (Halpern et al., 2022 [↗](#)) and high-throughput applications (Li et al., 2019 [↗](#); Walzik et al., 2015 [↗](#)).

On the other hand, another group of investigators tried to provide extremely cost-effective microscopy solutions that are accessible to everyone. The result are solutions that can be useful in detecting e.g. parasites in inaccessible regions of the world for simple diagnosis (Li et al., 2019 [↗](#)), but are incompatible with quantitative scientific imaging (Chagas et al., 2017 [↗](#); Cybulski et al., 2014 [↗](#); Diederich et al., 2020 [↗](#), 2019b [↗](#); Sharkey et al., 2016 [↗](#); Vera et al., 2016 [↗](#)). Most of these solutions lack excitation and emission for fluorescence imaging.

Here, we propose an instrument filling the gap between the low-cost open-source instruments for the general public and high-quality fluorescence imaging, as required for scientific research in molecular cell biology. We provide an automated high-resolution microscope with single molecule sensitivity that can be fully assembled and steered by lay users. Our system is based on the UC2 toolbox (Diederich et al., 2020 [↗](#), 2019a [↗](#); Wang et al., 2022 [↗](#)) using common optical elements, affordable electronic components and 3D printed parts and is combined with a high NA objective, motorized stage and intuitive GUI-based operation.

The result is a cost efficient and user-friendly high-resolution fluorescence microscope that is compatible with long-term live-cell imaging, single particle tracking and single molecule localization microscopy. All designs for the 3D parts, software and electronics are open-source (**Table 1** [↗](#)). We here describe microscope assembly, benchmarking and a number of high-resolution microscopy assays performed with our microscope.

Repository	Link:
General openUC2	https://github.com/openUC2/
Specific to this setup: UC.STORM	https://github.com/openUC2/UC2-STORM-and-Fluorescence
ImSwitch (UC2 specific)	https://github.com/openUC2/ImSwitch/ .
UC2-Rest	https://github.com/openUC2/UC2-REST
UC2-ESP32	https://github.com/youseetoo/uc2-esp32
UC2-ESP32 Firmware Flashintool (Online)	https://youseetoo.github.io/

Table 1

Github repositories for hardware, electronics and software.

Results

Assembly of a high-resolution UC2 microscope

We aimed to generate a cost-effective, high-resolution widefield fluorescence microscope that allows us to perform a variety of common cell biological imaging experiments. These include standard high-resolution diffraction limited immunofluorescence imaging, long-term live-cell imaging, single particle tracking and super-resolution microscopy. To fulfill these criteria, we reasoned that our system should have: i) excitation and detection powerful and sensitive enough for single molecule localization, thus requiring a laser and an appropriate detector. ii) capability to operate inside an incubator for long-term live-cell imaging, thus requiring small size and automation. iii) an intuitive and user-friendly GUI for maximal accessibility. Lastly, the entire system including the physical microscope, the electronics and the software must be open-source and be possible to assemble and operate with moderate technical know-how (**Figure 1a** [↗](#)). We reasoned that the quality of the objective is paramount to image quality, while most other optical parts of a light microscope can be mass-produced components. Furthermore, such a high-quality objective allows for the swift diagnosis of problems in the assembly of the optical path due to its high resolution and minimal chromatic and spatial aberrations. We decided to use an Olympus 1.49 NA oil objective, but any high-quality objective can be incorporated in our system, by adapting the objective mount to the requirements of the objectives' manufacturer.

Based on the guidelines described above and to ensure adaptability and modularity, we decided to build our open-source automated microscope on the basis of the UC2 toolbox using its injection molded cubes for the structural assembly. For ease of usage, we used an inverted configuration and assembled the microscope in three layers to keep the size and design compact (**Figure 1b** [↗](#)).

The high-precision x-y sample table and z-stage (**Figure 1c** [↗](#)) were low-cost commercial solutions that we motorized and implemented into the open-source, python-based microscope GUI as automated elements of the microscope. The emission pathway at the bottom of the microscope assembly is based on an Alvium 1800 U-158c CMOS camera from Allied Vision, which compared to conventional scientific cameras is more affordable by a factor of 30 to 60 times at the time of submission (**Figure 1d** [↗](#)), more compact and consumes less energy. One important point here is that the camera needs to be monochromatic as the Bayer pattern on a polychromatic camera would significantly reduce the sampling as well as the quantum efficiency of a camera chip, which is already lower compared to that of scientific cameras. The camera was connected to the computer via USB 3.0, allowing fast data transfer and maximal compatibility. Between the emission pathway and the sample stage, the excitation pathway contained a low-cost 638 nm diode laser with an optional beam expander (**Figure 1e** [↗](#)), that depending on the illumination requirements could be inserted or removed, modulating the illumination density and homogeneity of the excitation field (see **Figure 2c** [↗](#)). To trigger the laser, control laser power and to steer the x, y and z motors for focus and sample translation, we used a low-cost Arduino-based board that can be connected via USB3.0 to any computer. Electronics on the board and the USB-camera are then directly controlled via an adapted GUI of the python-based open-source software ImSwitch ([Moreno et al. 2021](#) [↗](#)), resulting in a fully automated microscope. Additionally, ImSwitch includes the "ImScript" an automation feature, which allows to execute a sequence of written *Python* commands with user defined parameters e.g. overnight time-series imaging experiments.

Furthermore, we took advantage of the open-source (OS) nature of ImSwitch and implemented the recently presented OS SMLM algorithms microEye ([Alsamsam et al., 2022](#) [↗](#)) to realize an online-running localization framework that gives instant qualitative feedback if the sample preparation works as expected. In addition, the feedback of these online-running image processing algorithms

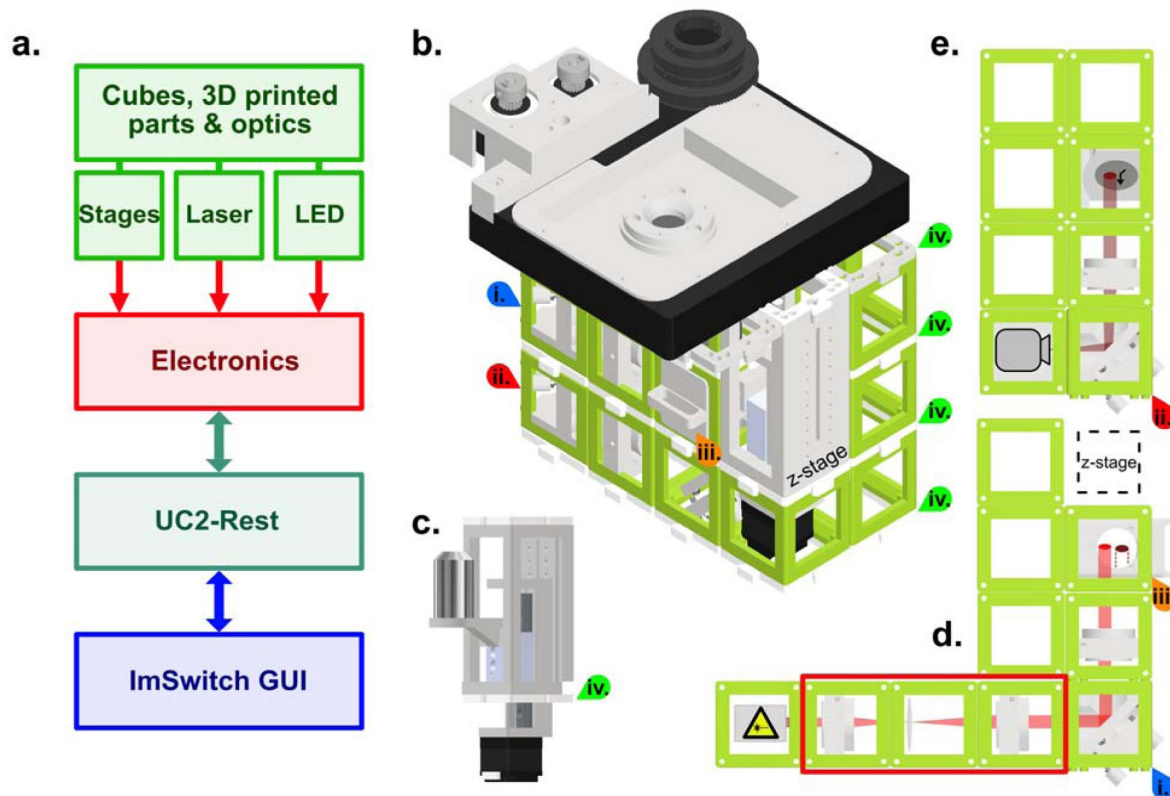


Figure 1

Schematic of the high-resolution-UC2 widefield microscope.

a. Overview of the different component categories of the microscope, from the hardware, through the electronics controlling, to the software and the ImSwitch-based GUI. **b.** Schematic of the complete setup. Red and blue arrows (**i.** and **ii.**) are references to help visualize the 3D structure. Top layer is the commercially bought XY-stage that was motorized for this project. Timing belts convert motor torque (two grey gears top left) into stage motion (bigger black gear middle) but have been omitted in this representation for clarity. Sample holder can be printed according to the sample used, circular in this case. **c.** Commercially bought precision Z-stage and high-NA objective (Olympus 60x/1.49 NA TIRF). **d.** Excitation layer. Laser emission (bright red light-path) is focused by a $f = 100$ mm lens onto the back focal plane of the objective. Orange arrow (**iii.**) marks the filter cube which has an excitation filter (Chroma ZET635/20x EX), a dichroic mirror (Chroma ZT640rdc) and on the bottom an emission filter (Chroma ET655lp long-pass for $\lambda > 655$ nm) to separate excitation from fluorescence. Red rectangle shows a telescope build including a diffusor (rotating piece of cling foil) in the focus point between both lenses. Telescope build magnifies and homogenizes the laser beam and can be used optionally. **e.** Detection layer, corresponds to the bottom layer in the setup (red star as reference). Emitted fluorescence (dark red light-path) is depicted with a $f = 100$ mm lens on the detector (Alvium 1800 U-158c CMOS camera from Allied Vision). All active optical elements, e.g. mirrors and lenses are mounted into the injection moulded cubes (50x50x50 mm, green) via custom 3D printed mounts. Cubes are connected laterally and vertically via the puzzle shaped connection pieces. Green arrows (**iv.**) indicate the connecting puzzle layers between the cubes.

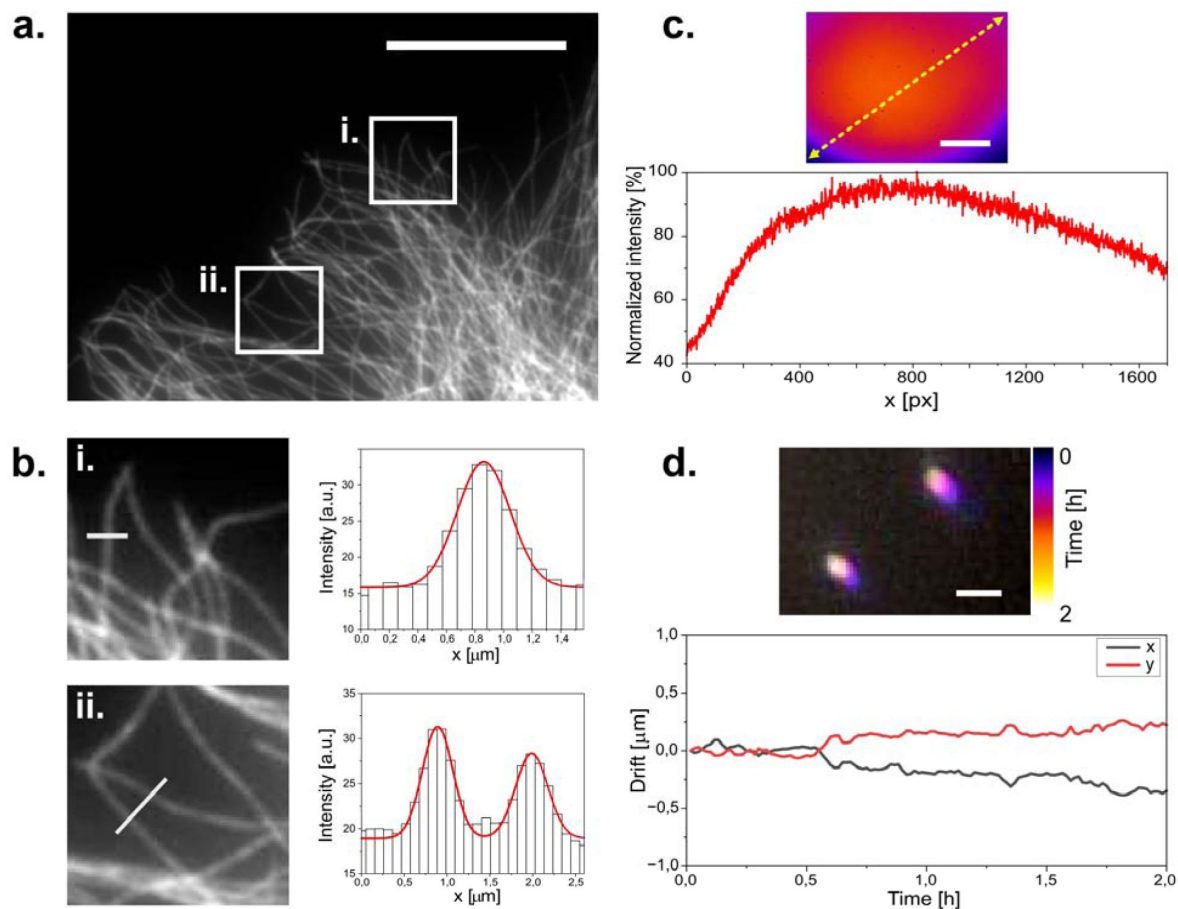


Figure 2

Widefield imaging of immuno-stained CV1 cells against tubulin (α and β -tubulin).

a. Widefield image of a tubulin stained CV1 cell. Two regions of interest are highlighted. Scale bar represents 20 μm . **b.** Zoomed in images of the selected ROIs in **a.** with respective microtubule profile plotted across the white line in the ROI images. ROI dimensions are 8x8 μm . Grey values are plotted as columns and fitted with gaussian functions (red line). Microtubule diameter can be estimated to 340 to 420 nm according to the full width at half maximum of the fits. **c.** Illumination profile of whole camera chip (1456x1088 px) at the sample plane as imaged on a far-red fluorescent Chroma slide. Scale bar represents 35 μm . **d.** Tetra-speck beads imaged over a period of three hours and color coded over time. Scale bar represents 1 μm . Bead positions are localized over time to measure their displacement to their initial position over time.

can be looped back into the hardware control to eventually compensate possible drift (e.g. autofocus). A large variety of available Napari (Ahlers et al., 2023 [↗](#)) plugins can be used to directly work on the data without moving them to a different computer.

All steerable electronics could alternatively be controlled with a PS4 controller for increased user-friendliness. The assembly can be housed with 3D printed walls or even black cardboard to block ambient light and increase laser safety. In [table 1](#) [↗](#), a list of the different microscope and software elements and the repository addresses to the corresponding materials and methods is displayed. A comprehensive online repository provides the bill of materials (BOM), as well as a set of instructions on how to setup the soft- and hardware ([Table 1](#) [↗](#)).

The 638 nm Red Laser Module used for illumination induced a recurrent pattern in the detection path when imaging a coverslip homogeneously covered with fluorescent molecules. Spectrometric measurement of the excitation laser beam confirmed that our low-cost laser had minor bands in the wavelength range of 650 - 670 nm, which we could remove using an additional band pass excitation filter (ZET635/20X). Our finding encouraged us in our approach to mix low-budget components with high quality optics. Focusing and positioning of the sample is done with commercial low-budget stages that offer great long-term stability. To render all three axes of movement computer controlled, we first motorized an x-y stage via torque transmission from the motor to the micrometer screws of the stage via timing belts. Secondly, we incorporated a motorized lead screw-driven (NEMA11, 50mm, Amazon, 50€) z-stage holding the objective into the modular design of the microscope.

Imaging experiments

We first used our microscope to image a sample of fixed mammalian cells stained against tubulin via immunofluorescence using AF647. When we imaged these cells with moderate laser intensity (35 W/cm^2), beam expander and 50 ms integration time, we found that cells were evenly illuminated ([Figure 2a](#) [↗](#)) and microtubules appeared as diffraction-limited lines inside cells ([Figure 2b](#) [↗](#)). To quantify flatness of illumination, we made use of a red auto-fluorescent benchmarking slide (Chroma). The measured fluorescence intensity was quite homogenous across the field of view, with variations of up to 25% of the maximal fluorescence intensity, excluding the bottom left and right corners ([Figure 2c](#) [↗](#)).

These conditions are compatible with high quality fluorescence microscopy in life science research. To test the stability of our system, we then imaged sub-diffraction fluorescent beads repeatedly over time. When we then calculated the displacement of the beads over time, we consistently found drift of less than $1 \mu\text{m}$ over 2 hours ([Figure 2d](#) [↗](#)). We concluded that our microscope may be suitable for imaging experiments that require repeated measurements of the same field of view such as live-cell imaging. We thus mounted live cells at room temperature on the optical table on our microscope and stained them with SiR-actin (Spirochrome). When we then repeatedly illuminated the cells with moderate laser power (18 W/cm^2) to excite SiR-actin fluorescence and with white light in transmission mode to image the cells, we found that the cells tolerated acquisition conditions for several hours without damage visible in fluorescence or widefield image ([Figure 3](#) [↗](#)). As expected, drift was negligible. Importantly, we found similar stability against drift over time, when we performed this experiment not on an optical table, but a wetlab bench over a layer of foam material (made of Polyethylene or alternatively Polyurethane) as a damping layer. This suggested that our microscope might be suitable for long-term imaging in an environment that does not provide vibration damping, such as a cell culture incubator.

Given that our microscope assembly is of small size, and connected via two flexible USB cables, we then moved on to perform live-cell imaging over longer periods in a cell culture incubator.

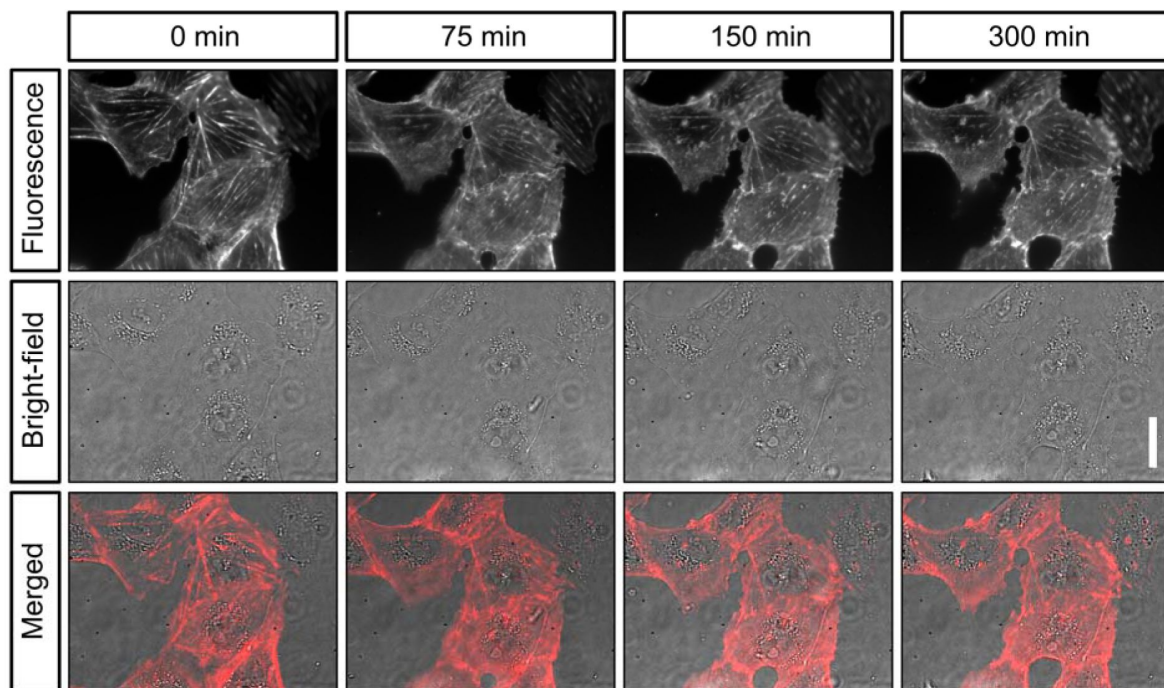


Figure 3

Live-cell imaging of actin in cultured cells.

Shown are still images of the same CV-1 fibroblast cells in SiR-actin fluorescence and widefield (background corrected) at different timepoints over 5 hours. Scalebar on the right represents 30 μm .

To do so, we used a standard 37 °C CO₂ incubator. The longer USB cables connecting the microscope to the exterior allowed for steering from the outside. When we plated live T98G cancer cells on coverslips and mounted them on the microscope sample holder in warm medium inside the incubator, they exhibited normal morphology and growth over many hours. When we performed a long-term imaging experiment using Syto deep-red fluorescence staining of the nucleus (Thermo Fisher), we found that cells kept dividing and moving on the coverslip for several days (**Figure 4**). We concluded that our setup was compatible with long-term live-cell imaging of cells.

Next, we aimed to further explore the single molecule capabilities of our microscope and performed single particle tracking (SPT) experiments on live CV-1 African green monkey cells stably expressing GPI-GFP. When we incubated these cells with streptavidin 655 nm quantum dots (QDs) coupled to biotin-tagged anti-GFP nanobodies (**Figure 5a**), we detected QDs readily as single diffraction limited entities on cells (**Figure 5b**). When we then localized emission centers and tracked the QDs over time, we could generate trajectories over many frames (**Figure 5c,d**) and extract parameters describing particle movement such as the diffusion coefficient (*D*, **Figure 5e**) and the moment scaling spectrum (MSS, **Figure 5f** (Ewers et al., 2005)). The MSS introduces a parameter that allows to quantitatively describe the type of motion a particle undergoes (MSS = 0 being immobility, < 0,5 Brownian diffusion and > 0,5 super-diffusive). Our measurements were in agreement with our previous observations (J. H. Li et al., 2020).

Discussion

Here we developed a cost effective, high-quality automated microscope that can be assembled by laypeople with minimal background knowledge in optics, electronics and informatics. The assembly guidelines including CAD files of the 3D printed parts, list of used components and software are openly accessible on the Github repositories (see **table 1**), thus offering maximal availability.

Our microscope is based on a simple modular assembly system called UC2 (You.See.Too., (Diederich et al., 2019a)). This framework is based on modular cubes that can hold different optical components in place and allow for simplified alignment and assembly. This flexibility furthermore facilitates the replacement of any lens, filter, laser camera or other components, allowing the user to implement already available parts into their setup, or to further trim down the setup by removing unused parts. We made use of an improved version where the cubes are made through injection molding, which provides a significantly higher manufacturing precision than the same cubes fabricated using 3D-printing, resulting in greater mechanical stability and reproducibility, while being available at low cost through mass production (Wang et al. 2022).

The hereby provided increased structural integrity makes the setup portable allows it to be reproducibly built and dismantled within hours. This facilitates its use in remote areas. Improved versions or updated parts could easily be shared digitally and printed locally, further promoting the advantage of digital manufacturing in the laboratory. The here created assembly can easily be replicated multiple times to increase imaging throughput still for a fraction of the cost of a single high-end microscope.

We present data in standard immunofluorescence microscopy, live-cell microscopy and long-term live-cell microscopy assays performed inside an incubator. Furthermore, we show single particle tracking data of semiconductor quantum dots on live cells and single molecule localization based super-resolution microscopy. In all cases, the quality of data we generated is comparable to those generated by systems orders of magnitude more expensive.

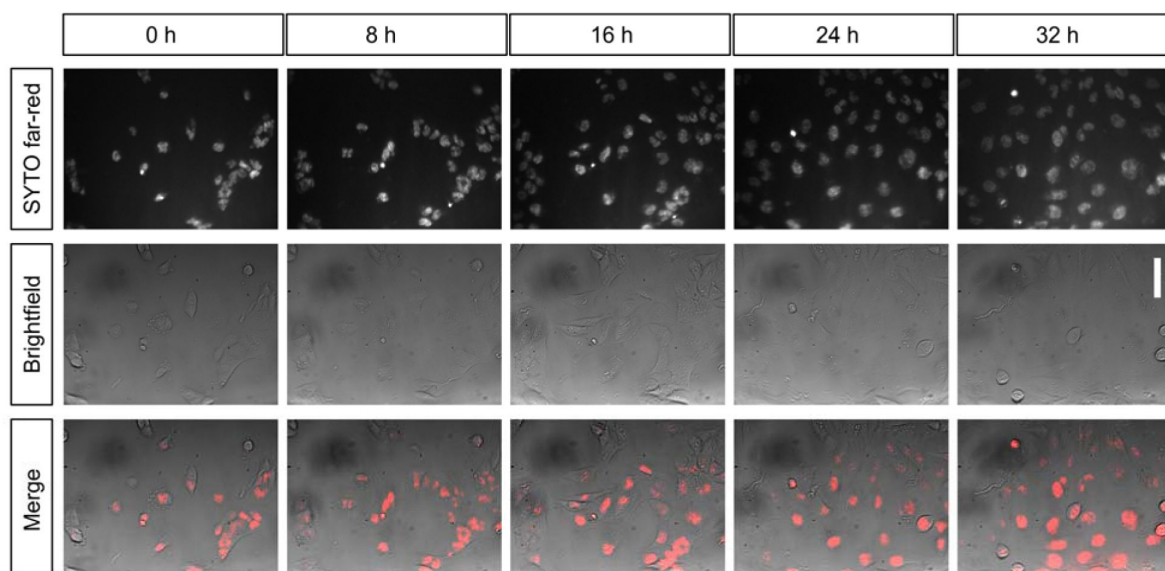


Figure 4

Long-term live-cell imaging of cultured cells inside the incubator.

Shown are still images of the same population of T98G glioblastoma cells over time in the same field of view. Cells are stained with SYTO nucleic-acid fluorescent dye and imaged in widefield (background corrected) every 20 minutes for over 32 hours inside an incubator. Scalebar on the right represents 30 μm .

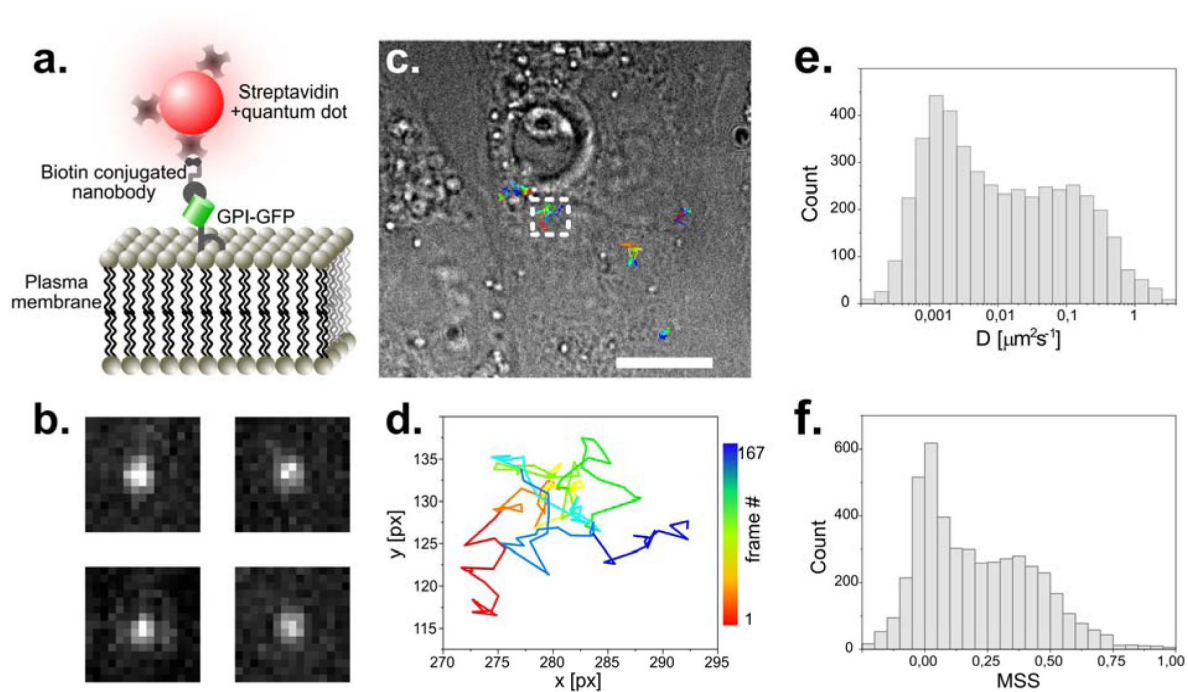


Figure 5

Single particle tracking of GPI-GFP in live CV1 cells using functionalized Quantum dots.

a. Schematic of the construct used to track single GPI- molecules with quantum dots. **b.** Exemplary raw images of single fluorescent Quantum Dots. Image dimension is $1.6 \mu\text{m} \times 1.6 \mu\text{m}$. **c.** Trajectories of several single molecules visualized on a top-light illuminated image of CV1 cells. Scale bar represents $10 \mu\text{m}$. **d.** Selected trajectory of a single particle over 167 time-points. Sampling rate is 25 ms, pixel size 104 nm. **e.** Histogram of the diffusion coefficient of around 5000 tracks with 20 or more detected time points each. **f.** MSS (moment scaling spectrum) of the same population as plotted in **e**. Since we were readily able to detect single quantum dots, which are single emitters of few nanometers in size, we reasoned, it might be possible to perform single molecule localization microscopy (SMLM) experiments as well. We immuno-stained CV-1 cells against tubulin using AF647 coupled secondary antibodies and incubated coverslips in BME/GLOX buffer for effective pumping into the dark state. When we then illuminated our sample with high intensity laser light (in our case $\sim 1\text{ kW/cm}^2$), we found that the intense staining quickly faded and gave way to the typically observed blinking in SMLM experiments. We took a timeseries of 30'000 frames and localized the detected emitters using *Thunderstorm* (Ovesný et al., 2014). Even though we were limited to widefield illumination, we achieved a drastic improvement in resolution in reconstructed SMLM micrographs (Figure 6a). When we measured the profile of individual microtubules, we could easily detect the typical “railroad track” pattern with a spacing of 38 – 43 nm, indicative of high quality SMLM imaging (Figure 6b,c). To quantify the achieved resolution, we used Fourier ring correlation (Nieuwenhuizen et al., 2013) and found that the overall achieved resolution was 93 nm, clearly below the optical resolution limit (Figure 6d).

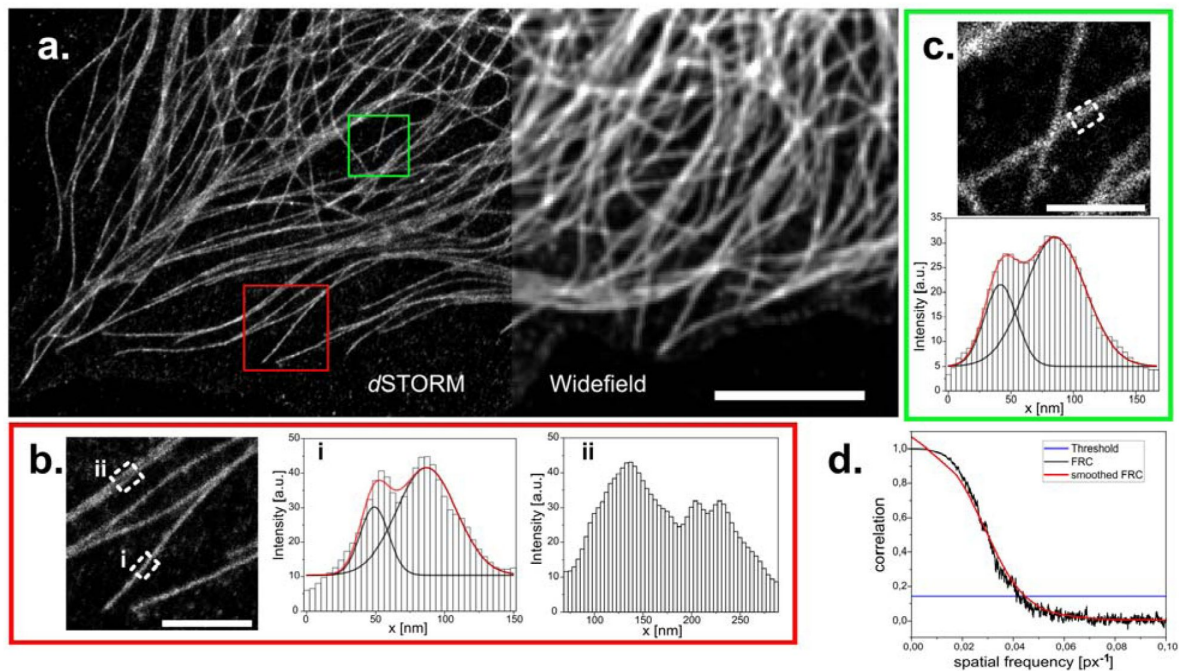


Figure 6

dSTORM reconstruction of immuno-stained CV1 cells against tubulin (a and γ -tubulin).

a. Widefield image and reconstructed super-resolution image of 30'000 widefield frames, localized and reconstructed using *ThunderSTORM*, applied parameters described in the methods part. Reconstruction is displayed with the average shifted histogram method, including a 25-fold magnification. Scale bar represents 5 μm . Two regions of interest in the reconstruction are marked (red square for **b.** and green square for **c.**) and zoomed in (scale bar 1 μm). Profiles within these ROIs are measured by averaging the profile of the reconstructed image over 250 nm along the microtubule (white dashed rectangles). White histogram bars are the intensity values of the pixels in the reconstruction. Two peaks arise from the circular shape of microtubules. These were analyzed with a double gaussian fit (red curves are the cumulated single black fits). Distances of (38 ± 2) nm (red box) and (43 ± 2) nm (green box) could be extrapolated. **d.** Fourier ring correlation of the complete reconstructed image. Threshold is left to the preset 1/7. Overall resolution is estimated at 93 nm.

Our system thus allows for sophisticated imaging assays to be performed at a professional level with greatly reduced cost. At the same time, electronic control of components and remote steering of the microscope can be implemented without background knowledge in electronics. This makes the generation of high-quality image data accessible especially for life science research groups with limited monetary means also in lower income countries. The main limitation of our system in this direction is the requirement of a high-NA objective, the by far most expensive component of the setup. Since for scientific imaging, high quality data and thus high-NA and corrected objectives are essential, this component cannot be replaced. However, the quality of low-cost objectives has recently been improving, even with objectives allowing for effective single molecule detection. The reduced NA diminishes the localization precision and the signal to noise ratio, thus limiting the possible applications. Development of brighter dyes might also be an alternative to improve the imaging quality enough to account for low budget lasers and filters.

Due to its modular assembly, extensions may be easily implemented to our microscope. Depending on the specific research aim, multiple excitation and emission wavelengths may be desirable, but would require a redesign of the frame to incorporate motorized filter switching and more light sources. Alternatively, multiband filters in combination with fiber-coupled and thus combinable laser sources are also possible. In the future, the possibilities of this system could be significantly expanded by adding more modules specific for additional capabilities and we specifically invite the community to do so. Especially the combination with a low-budget spinning disc (Halpern et al., 2022 [↗](#)), TIRF illumination or light-sheet imaging (Diederich et al., 2020 [↗](#)) are possible future applications.

Further steps towards user-friendliness include the replacement of the laser light-source by a high-power LED to reduce the requirement of laser safety. Another step would be to render the electronics assembly fully plugable, thus eliminating the soldering steps that are required in the current assembly.

Besides the application in basic research, this cheap and easy to assemble setup would be useful for teaching the next generation of scientists that more than ever have to be able to synthesize capabilities spanning several disciplines. We endorse making scientific instruments available for more scientists, students, as well as the general public. Our system builds on efforts in open science regarding software (ImSwitch, (Moreno et al., 2021 [↗](#)) and microscope hardware (UC2 (Diederich et al., 2019a [↗](#))) while benefitting from a growing field of open-source microscopy and laboratory hardware in general (Almada et al., 2019 [↗](#); Auer et al., 2018 [↗](#); Hohlbein et al., 2022 [↗](#); Martens et al., 2019 [↗](#); Wenzel, 2023 [↗](#)). Taken together with efforts in making fluorescent dyes available (Lavis, 2021 [↗](#)), fundamental obstacles to the free diffusion of capabilities and know-how in science are being removed. We hope that our live-cell high-resolution incubator microscope (U.C.STORM) will catalyze these efforts.

Methods

Component fabrication

In the components list, a detailed table describes all the different pieces used for the microscope, including the price of each component. The CAD designs as well as further information about UC2 or setup building can be found on the Github repository (see **table 1** [↗](#)). The XY-stage was a manual commercial model that was modified and motorized for the application. The Z stage was also commercially bought, but already manufactured with a motor. Both stages were chosen because of their low price and convenient use. Custom-made stage solutions to print and build can be found on the UC2 repository (**Table 1** [↗](#)). In the slicing software Prusa slicer, the original Prusa i3 MK3S & MK3S+ printer was set. The parts were printed with the 0.15 mm SPEED or 0.15 mm QUALITY settings. The two materials used in this project were the Prusament PLA (Polylactic Acid)

and Prusament ABS (Acrylnitril-Butadien-Styrol) filaments (both from Prusa), which can both be selected in the slicing software ABS was used wherever heavy loads may deform the setup in the event of temperature gradients (e.g. objective mount). Materials were printed with PLA infill (20 - 40%) ($T_{\text{print}} = 215\text{ }^{\circ}\text{C}$; $T_{\text{bed}} = 60\text{ }^{\circ}\text{C}$). Materials printed with ABS infill (20 - 40%) ($T_{\text{print}} = 255\text{ }^{\circ}\text{C}$; $T_{\text{bed}} = 100\text{ }^{\circ}\text{C}$).

UC2 setup assembly

In the UC2 system, optical components were mounted in standardized cubes. To make alignment possible over an optical excitation or emission path over several cubes distance, the components must experience a large enough degree of freedom within the cubes so they can be moved manually in angle and exact position, but low enough to keep components immobile over prolonged time periods. The challenge presented by tilted optics can on the one side be addressed during the design process; broader mounts/adapters are less prone to tilt due to higher contact surface with the cube. Once the optics were orthogonal to the light path, their position in relation to the optical axis was set manually with highest possible accuracy, using appropriate mounts.

The fine adjustment was done with tiltable mirrors, as using translational mounts for every optical element would be complex to produce or costly. A helpful tool for adjustment was to position an aperture in the excitation pathway. The maximum of the laser spot was set to be in the center of the aperture hole and was adjusted using the mirror. The emission path was adapted in the same way, as imaging a fluorescent surface allows to position the maximal laser intensity spot on the center of the detector field of view (also with a mirror).

The setup entails two imaging channels: the laser induced fluorescence (excitation path) and the LED top-light illumination (650 nm Star LED, Cree, 3W, brightfield). The top-light light source is mechanically detangled from the stages to remain unaltered by sample motion. As the filter cube in the microscope is immobile, the top-light illumination requires either a high-power white LED or an LED that emits light in a wavelength range that is not blocked by the dichromatic mirror or the emission filter (in our case 650-660 nm).

Setup parts availability and price

The different 3D printed parts, electronics and optics used in the assembly, the corresponding CAD files as well as an estimation of the price can be found in the corresponding Github repositories, as listed in **table 1** [↗](#). General information about the UC2 system can be found on the UC2 repository.

Image acquisition and control software

For the microscopy control, acquisition and reconstruction software, we chose ImSwitch because its modular design adapts very well to the inherently modular nature of the cube-based optics platform. A novel firmware “UC2-ESP32” was developed, which is fully integrated into the python-based ImSwitch structure. Using a unified REST API-like interface, lasers, motors, LEDs, LED arrays and readout sensors can be controlled serially via WiFi and USB. To ensure wide distribution of this device adapter, we used a readily available hardware module that can control multiple stepper motors and easily adapt other hardware components such as the LEDs and lasers. An online-based flash tool loads the latest firmware onto the Wemos ESP32 D1 R32 in combination with the CNC Shield to create a fully functional microscopy control unit. Detailed documentation can be found in the online documentation (<https://youseetoo.github.io/↗>). A Python library “UC2-REST” adapts the USB serial and integrates it into the ImSwitch software. In this project, the strengths of open-source were fully unleashed by extending the control and acquisition algorithm ImSwitch with an online SMLM reconstruction algorithm by (Alsamsam et al., 2022 [↗](#)). The user can select different parameters from the MicroEye framework to locate blinking events directly in the Napari viewer (Sofroniew et al., 2022 [↗](#)) to directly track possible reconstruction or sample artefacts. The UC2-specific fork amongst other repositories can be found in **Table 1** [↗](#).

Imaging modalities

Illumination

The illumination is provided by the 638nm Red Laser Module 500mW Round Dot Focusable TTL 3050 (acquired on lasernets.com). When implementing the telescope and beam homogenizer, the illumination density reaches up to 90 W/cm^2 but could be tuned to any required value below that. In the focal point between both lenses a rotating piece of cling film diffuses the beam. The variance in the laser density over the field of view excluding the corners was measured with a fluorescent slide and is reproducibly lower than 25%.

Additionally, for long term experiments, lacking fluorescence intensity can to an extent be compensated with longer detector exposure times. To perform SMLM i.e. *d*STORM in our case, the telescope is removed to have maximal laser density. The total available power at the sample plane is 83 mW thus a laser density of around 520 W/cm^2 . But since the laser spot is smaller than the FOV, sufficient laser density can be reached when only using a subset of the FOV (approximately a third of the FOV) can be used with sufficient laser power (over 970 W/cm^2) and a variation in laser intensity of about 25% of the laser intensity maximum.

Objectives and pixel sizes

For the widefield imaging, the room temperature live cell imaging, the single particle tracking as well as the *d*STORM experiment, a 60x Olympus 1.49NA oil immersion objective (UPLAPO60XOHR) is used. The setup optics have accordingly been chosen for maximal resolution thus matching the Nyquist sampling rate on the camera. The pixel-size is determined with a calibrated grid and a value of 104.1 nm is extrapolated. For the incubator measurements, a 20x Olympus 0.6 NA air (MXPLFLN20X) is used. The optics in the setup are not adjusted to this objective, resulting in a pixel-size of 319.4 nm.

Coverslip and cell seeding

Coverslip preparation

Metal holders were used in combination with circular coverslips (25 mm) which were first cleaned with a 2% Hellmanex solution for 10 minutes then cleaned in ethanol for 10 minutes and were finally placed in the plasma cleaner for 15 minutes after having dried. The 8-well sample-holders (Ibidi) were washed with 1 M KOH for 10 min then rinsed with H_2O and dried.

Cell culture medium

Cell culture medium was on a phenol red free DMEM base with 1% GlutaMAX and 10% FBS.

Cell seeding

Cells were detached from the cell culture dish with a 2 mM EDTA solution. 30'000 cells were seeded on a 25 mm circular coverslip for *d*STORM experiments and general wide- field imaging. For the live cell experiments, 55'000 to 80'000 cells were seeded onto 25 mm coverslips, alternatively 15'000 - 25'000 cells per well when using an 8 well chamber.

Microtubule staining

CV-1 cells (Wild-type *Cercopithecus aethiops* kidney fibroblasts) were first washed with 37 °C warm PEM buffer. The cells were fixed with warm PEM with 4% PFA, 0.05% GA and 0.1% TX 100 for 20 minutes. After fixation, sample was quenched in 8 minutes with NH_4Cl /PBS (50 mM) then

washed 3 times with PEM buffer. The cells were then permeabilized with 0.3 % TX100 in PEM for 5 minutes and washed with PEM buffer. The sample was then blocked for 30 minutes in imageIT followed by 1 hour in a 4 % HS, 1 % BSA, 0.1 % TX100 PEM based solution.

The sample was incubated overnight (12Lhours) at 4 °C with a primary mouse anti alpha- and beta- tubulin IgG antibodies (Sigma-Aldrich (REF: T5168) and Sigma-Aldrich (REF: T5293)) diluted by a factor of 1 to 350 in a 1% BSA and 0.1 % TX100 PEM based solution. Sample was thoroughly washed 3 times for 5 minutes with a 1 % BSA and 0.1 % TX100 PEM-based solution. The cells were then incubated with the secondary donkey anti-mouse antibody conjugated with the AF647 fluorophore, diluted by a factor of 1 to 250 in a 1% BSA and 0.1 % TX100 PEM based solution at room temperature for 2 hours. The 3 washing steps were then repeated, as previously mentioned for the primary antibody.

The sample was the post-fixated with a 4% PFA diluted in PEM solution, for 15 minutes. The cells were then quenched with a 50 mM NH₄Cl solution diluted in PBS for 8 minutes. Finally, the cells was washed 3 times.

Single particle tracking

Anti-GFP nanobodies (LaG-16 anti-GFP, own production according to (Fabricius et al., 2018 [link](#))) were added to a biotin (α -Biotin- ω -(succinimidyl propionate)-24(ethylene glycol)) solution at double molar excess. After 1 hour of incubation on a shaker (300 rpm), the nanobody-biotin construct was purified by sequential filtration through three 7 kDa MWCO desalting columns.

The biotin conjugated anti-GFP nanobodies were mixed in a 1 to 1 ratio with the streptavidin coated quantum dots and incubated for 10 minutes at room temperature. The solution was then diluted into 1 ml of live cell medium. The cells were then incubated at room temperature with this solution for 15 minutes. Afterwards the supernatant was removed and replaced with regular live cell-medium (Thermo Fischer).

SiR actin staining

The CV-1 cells were washed in 37° C warm Medium (DMEM without phenol red + FBS + glutamax). 1 μ l of the original SiR Actin (Spirochrome, 100 nM) solution was diluted in 1 ml of 37° C warm Medium (DMEM without phenol red + 10 % FBS + 1 % glutamax). To this solution, 1 μ l of verapamil was added. The sample medium was removed and exchanged with this solution. After 1 hour of incubation in the incubator, the sample was imaged in a live cell imaging solution (Invitrogen) for the following hours at room temperature.

Nucleic staining

T98G cells were washed in 37°LC prewarmed medium (DMEM without phenol red + 10 % FBS + 1 % glutamax). For staining, 1L μ l of SYTO™ Red Fluorescent Nucleic Acid Stain (Invitrogen) was diluted in 1Lml of complete medium and incubated with the sample for 30 minutes at 37°LC in the incubator, then imaged over prolonged time.

(d)STORM imaging

Imaging buffer

Imaging buffer was made of 150 mM Tris-HCl, 1.5 % β -Mercaptoethanol, 0.5 % (v/w) glucose, 0.25 mg/ml glucose oxidase and 20 μ g/ml catalase (pH 8.8).

Image acquisition

The 30'000 frames of the (d)STORM data set have been acquired with ImSwitch on the full camera chip (1456 x 1088 px). The exposure time was 20 ms, the gain was set to 20 (maximal value), the black-level and offset to 0 and no time delay was set between two frames. Laser power was set to 1024 (maximal value). Setup was positioned on a regular table with a layer of foam material to damp vibrations. The room was left during the measurement.

The images were acquired within around 15 minutes and saved directly on the computer disc in a .hdf5 file format for time efficiency and data handling advantages.

Analysis and reconstruction

The (d)STORM data-set was imported as a whole into ImageJ as a hdf5 file. Acquiring the full CMOS chip 30'000 times generated enormous datasets (95.6 Gb). To load the file, a Fiji plugin to load the stack as virtual stack was used. (N5, (Saalfeld et al., 2022 [link](#))). The stack of images was then cropped to retain the region of interest, which has been positioned to match the region where sufficient laser density induces blinking of single molecules during the acquisition. The analysis was done using ThunderSTORM (Ovesný et al., 2014 [link](#)), a publicly available SMLM reconstruction plugin on ImageJ. The camera parameters mentioned in the image acquisition section were used for the analysis. Image filtering was done with a Difference- of-Gaussian filter $\sigma_1 = 1.1$ px; $\sigma_2 = 1.7$ px. The approximate localization of molecules was done using the local maximum method with default settings (std(Wave.F1) as peak intensity threshold and 8-neighborhood connectivity). Sub-pixel localization of molecules was done with the PSF: Integrated Gaussian (Fitting radius 3px) and maximum likelihood fitting method (initial sigma 1.7 px). Reconstruction of localized the raw-data was visualized using the average shifted histograms method (Magnification: 25x; lateral shifts: 3).

Author contributions

HE and AMV acquired funding and conceptualized the study. BD developed the UC2 platform and adapted the electronics and software. ACZ and BD build and improved the microscope and its application. ACZ conducted the experiments and data analysis. ACZ, HE and BD wrote the paper.

Acknowledgements

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Editors

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Reviewer #1 (Public Review):

The authors have developed an open-source high-resolution microscope that is easily accessible to scientists, students, and the general public. The microscope is specifically designed to work with incubators and can image cells in culture over long periods. The authors provide detailed instructions for building the microscope and the necessary software to run it using off-the-shelf components. The system has great potential for studying cell biology and various biological processes.

The authors' work will make scientific instruments more accessible and remove obstacles to the free diffusion of capabilities and know-how in science. This important contribution will enable more people to conduct scientific research.

<https://doi.org/10.7554/eLife.89826.2.sa1>

Reviewer #2 (Public Review):

Making state-of-the-art (super-resolution) microscopy widely available has been the subject of many publications in recent years as correctly referenced in the manuscript. By advocating the ideas of open-microscopy and trying to replace expensive, scientific-grade components such as lasers, cameras, objectives, and stages with cost-effective alternatives, interested researchers nowadays have a number of different frameworks to choose from. In the iteration of the theme presented here, the authors used the existing modular UC2 framework, which consists of 3D printable building blocks, and combined a cheapish laser, detector and x,y,(z) stage with expensive filters/dichroics and an expensive high-end objective (>15k Euros).

The choice of using the UC2 framework has the advantage, that the individual building blocks can be 3D printed, although it should be mentioned that the authors used injection-moulded blocks that will have a limited availability if not offered commercially by a third party. The

strength of the manuscript is the tight integration of the hardware and the software (namely the implementations of imSwitch as a GUI to control data acquisition, OS SMLM algorithms for fast sub-pixel localisation and access to Napari).

The presented experimental data is convincing, demonstrating (1) extended live cell imaging both using bright-field and fluorescence in the incubator, (2) single-particle tracking of quantum dots, and (3) and STORM measurements in cells stained against tubulin.

For the revised (current) version of the manuscript, the authors further polished the manuscript and, more importantly, added plenty of information on the GitHub page that should make it significantly easier for interested researchers to replicate the instrument.

Overall, this is compelling work that is helping to make super-resolved microscopy more accessible.

<https://doi.org/10.7554/eLife.89826.2.sa0>

Author Response

The following is the authors' response to the original reviews.

Reviewer #2 (Public Review):

Making state-of-the-art (super-resolution) microscopy widely available has been the subject of many publications in recent years as correctly referenced in the manuscript. By advocating the ideas of open-microscopy and trying to replace expensive, scientific-grade components such as lasers, cameras, objectives, and stages with cost-effective alternatives, interested researchers nowadays have a number of different frameworks to choose from. In the iteration of the theme presented here, the authors used the existing modular UC2 framework, which consists of 3D printable building blocks, and combined a cheapish laser, detector and x,y,(z) stage with expensive filters/dichroics and a very expensive high-end objective (>15k Euros). This particular choice raises a first technical question, to which extent a standard NA 1.3 oil immersion objective available for <1k would compare to the chosen NA 1.49 one.

Measurement of the illumination quality (e.g. the spectral purity) of low budget lasers convinced us of the necessity to use spectral filtering. These cannot be replaced with lower budget alternatives, to still retain the necessary sensitivity to image single molecules. As expected, the high-quality objectives are able to produce high-quality data. Lower budget alternatives (<500 €) to replace the objective have been tried out. Image quality is reduced but key features in fluorescent images can be identified (see figure S1). The usage of a low budget objective for SMLM imaging is possible, but quality benchmarks such as identifying railroad tracks along microtubule profiles is not possible. Their usage is not optimal for applications aiming to visualize single molecules and might find better application in teaching projects.

The choice of using the UC2 framework has the advantage, that the individual building blocks can be 3D printed, although it should be mentioned that the authors used injection-molded blocks that will have a limited availability if not offered commercially by a third party. The strength of the manuscript is the tight integration of the hardware and the software (namely the implementations of imSwitch as a GUI to control data acquisition, OS SMLM algorithms for fast sub-pixel localisation and access to Napari).

The injection-molded cubes can be acquired through the OpenUC2 platform. Alternatively, the 3D printable version of the cubes is freely available and just requires the user to have a 3D printer. https://github.com/openUC2/UC2-GIT/tree/master/CAD/CUBE_EmptyTemplate

The presented experimental data is convincing, demonstrating (1) extended live cell imaging both using bright-field and fluorescence in the incubator, (2) single-particle tracking of quantum dots, and (3) and STORM measurements in cells stained against tubulin. In the following I will raise two aspects that currently limit the clarity and the potential impact of the manuscript.

First, the manuscript would benefit from further refinement. Elements in Figure 1d/e are not described properly. Figure 2c is not described in the caption. GPI-GFP is not introduced. MMS (moment scaling spectrum) could benefit from a one sentence description of what it actually is. In Figure 6, the size of the STORM and wide-field field of views are vastly different, the distances between the peaks on the tubuli are given in micrometers rather than nanometers. (more in the section on recommendations for the author)

Second, and this is the main criticism at this point, is that although all the information and data is openly available, it seems very difficult to actually build the setup due to a lack of proper documentation (as of early July 2023).

1. The bill of materials (<https://github.com/openUC2/UC2-STORM-and-Fluorescence#bill-of-material>) should provide a link to the commercially available items. Some items are named in German. Maybe split the BoM in commercially available and 3D printable parts (I first missed the option to scroll horizontally).

1. The links to the XY and Z stage refer to the general overview site of the UC2 project (<https://github.com/openUC2/>) requiring a deep dive to find the actual information.

1. Detailed building instructions are unfortunately missing. How to assemble the cubes (pCad files showing exploded views, for example)? Trouble shooting?

1. Some of the hardware details (e.g. which laser was being used, lenses, etc) should be mentioned in the manuscript (or SI)

I fully understand that providing such level of detail is very time consuming, but I hope that the authors will be able to address these shortcomings.

1. The bill of materials has been and will also in future still be improved. The items have been sorted into UC2 printed parts and externally acquired parts. The combination of part name as well as provider enables users to find and acquire the same parts. Additionally, depending on the country where the user is located, different providers of a given part might be advantageous as delivery means and costs might vary.

2. The Z-stage now has a specific repository with different solutions, offering different solutions with different levels of movement precision. According to the user and their budget, different solutions can be optimal for the endeavor.

<https://github.com/openUC2/UC2-Zstage>

The XY stage now also has a detailed repository, as the motorizing of the stage requires a fair amount of tinkering. The video tutorials and the detailed instructions on stage motorizing should help any user to reproduce the stage shown within this manuscript. <https://github.com/openUC2/UC2-Motorized-XY-Table>

1. The updated repository has a short video showing the general assembly of the cubes and the layers. Additionally, figure S2 shows all the pieces that are included in every layer (as a photograph as well as CAD). An exploded view of the complete setup would certainly be a helpful visualization of the complete setup. We however hope that the presented assembly tutorials and documents are sufficient to successfully reproduce the U.C.STORM setup.

First, we want to thank the reviewers for their effort to help us improving our work. We apologize for any trivial mistakes we had overlooked. Please find below our answers to the very constructive and helpful comments of the editors.

Recommendations for the authors:

Reviewer #1 (Recommendations for The Authors):

To complement the current data set:

Figure 2(a & b): Panels i & ii, were chosen on the area where the distribution of the laser appears to be flatter. Can the authors select microtubules from a different section? Otherwise, it is reasonable to also crop the field-of-view along the flatter area (as done in Fig 6).

Figure 2 was changed to according to the reviewer's suggestions. The profiles of microtubules from a different section have similar profiles, but the region with best illumination thus best SNR of the profile have been used for the figure.

Figure 2(c): The current plot shows the gaussian distribution which does not appear to be centered. Instead of a horizontal line, can the authors provide a diagonal profile across the field of view and update the panel below?

A diagonal cross-section of the illuminated FOV is provided in figure 2 to replace the previous horizontal profile. The pattern seems not to be perfectly radially symmetric, and more light seems to be blocked at the bottom of the illumination pattern compared to the top. A possible improvement can be provided by a fiber-coupled laser, that could provide a more homogeneous illumination while being easier to handle in the assembly process.

Author response image 1.

Diagonal cross-section of the illuminated FOV. Pixel-size (104nm) is the same as in figure 2. Intensity has been normalized according to the maximal value.

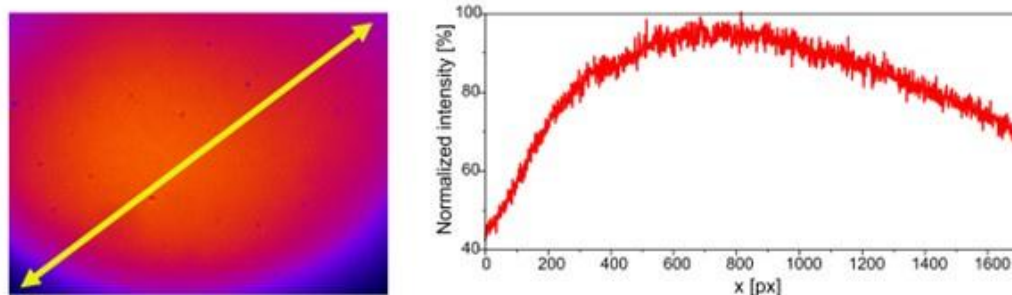


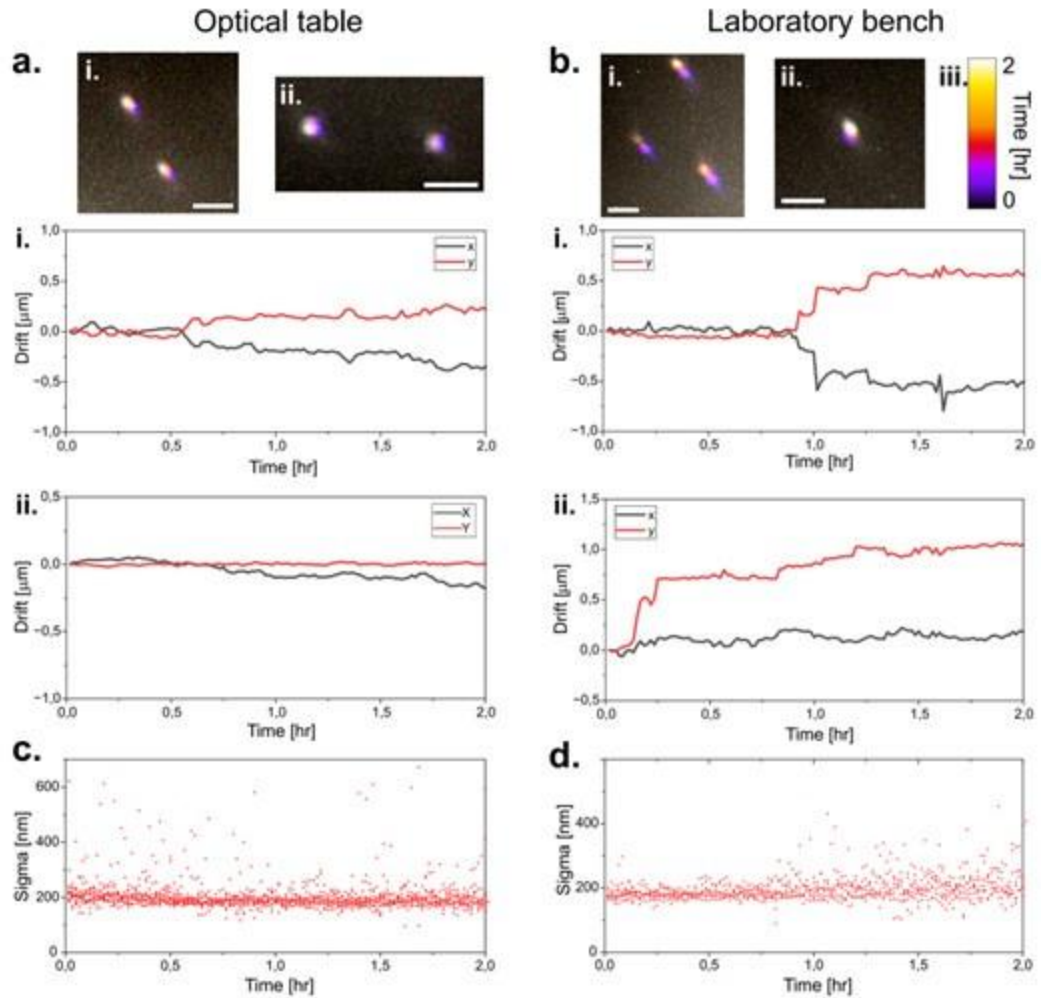
Figure 2(d): The system presents a XY drift of ~500nm over the course of a couple of hours. However, is not clear how the focus is being maintained. Can the authors clarify this point and add the axial drift to the plot?

The axial position of the sample could be maintained over a prolonged period of time without correcting for drift. Measurements where an axial shift was induced by tension pulses in the electronics have been discarded, but the stability of the stage seems to be sufficient to allow for imaging without lateral and axial drift correction. The XY drift measurement displayed in Figure 2(d) can be extended by measuring the σ of the PSF over time. The increase of σ would suggest an axial displacement in relation to the focus plane. In these measurements, a slight axial drift can be seen, the fluorescent beads however can still be localized over the whole course of the measurement.

A separate experiment was performed, using the same objective on the UC2 setup and on a high-quality setup equipped with a piezo actuator able to move in 10 nm steps. The precise Z steps of the piezo allows to reproducibly swipe through the PSF shape and to give an estimate of the axial displacement of the sample, according to the changes in PSF FWHM (Full Width at Half Maximum). When superimposing the graph with the UC2 measurement of fluorescent beads with the smallest possible Z step, an estimate about the relative axial position of the sample can be provided. The accuracy of the stage however remains limited.

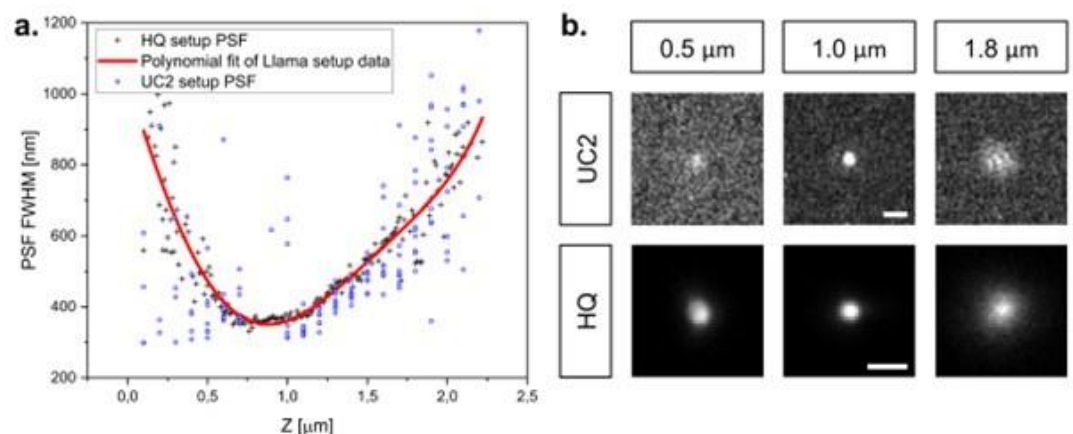
Author response image 2.

Drift Figure: a. Drift of fluorescent TS beads on the UC2 setup positioned upon an optical table over a duration of two hours. Beads are localized and resulting displacement in i. and ii. are plotted in the graphs below. The procedure is repeated in b. with the microscope placed on a laboratory bench instead. c. (for the optical table i.) and d. (for the laboratory bench i.) show the variation in the sigma value of the localized beads over the measurement duration. As the sigma values changes when the beads are out of focus, the stability of the setup can be confirmed, as it remains practically unchanged over the measurement duration.



Author response image 3.

Z-focus Figure: Estimation of the axial position of TS beads on the UC2 setup. a. The change in PSF FWHM was quantified by acquiring a Z stack of a beads sample. The homebuilt high-quality setup (HQ) was used as a reference, by using the same objective and TS sample. The PSF FWHM on the UC2 setup was measured using the lowest possible axial stage displacement. A Z-position can thus be estimated for single molecules, as displayed in b.



Addressing the seemingly correlated behavior of the X and Y drift:

Further measurement show less correlation between drift in X and in Y. Simultaneous motion in X and Y seems to indicate that the stage or the sample is tilted. The collective movement in X and Y seems accentuated by bigger jumps, probably originating from vibrations (as more predominantly shown in the measurements on the laboratory bench compared to the optical table). Tension fluctuations inducing motion of the stage are possible but are highly unlikely to have induced the drift in the displayed measurements.

Figure 3: Can the authors comment on the effect or otherwise potential effect of the incubator (humidity, condensation etc) may have on the system (e.g., camera, electronics etc)?

When moving the microscope into the incubator, the first precaution is to check if the used electronics are able to perform at 37° C. Then, placing the microscope inside the incubator can induce condensation of water droplets at the cold interfaces, potentially damaging the electronics or reducing imaging quality. This can be prevented by preheating the microscope in e.g. an incubator without humidity, for a few hours before placing it within the functional incubator. The used incubator should also be checked for air streams (to distribute the CO₂), and a direct exposure of the setup to the air stream should be prevented. The usage of a layer of foam material (e.g. Polyurethane) under the microscope helps to reduce possible effects of incubator vibrations on the microscope. The hydrophilic character of PLA makes its usage within the incubator challenging due to its reduced thermal stability. The temperature also inherently reduces the mechanical stability of 3D printed parts. Using a less hydrophilic and more thermally stable plastic, such as ABS, combined with a higher percentage of infill are the empirical solution to this challenge. Further options and designs to improve the usage of the microscope within the incubator are still in development.

Figure 5: Can the authors perform single molecule experiments with an alternative tag such as Alexa647?

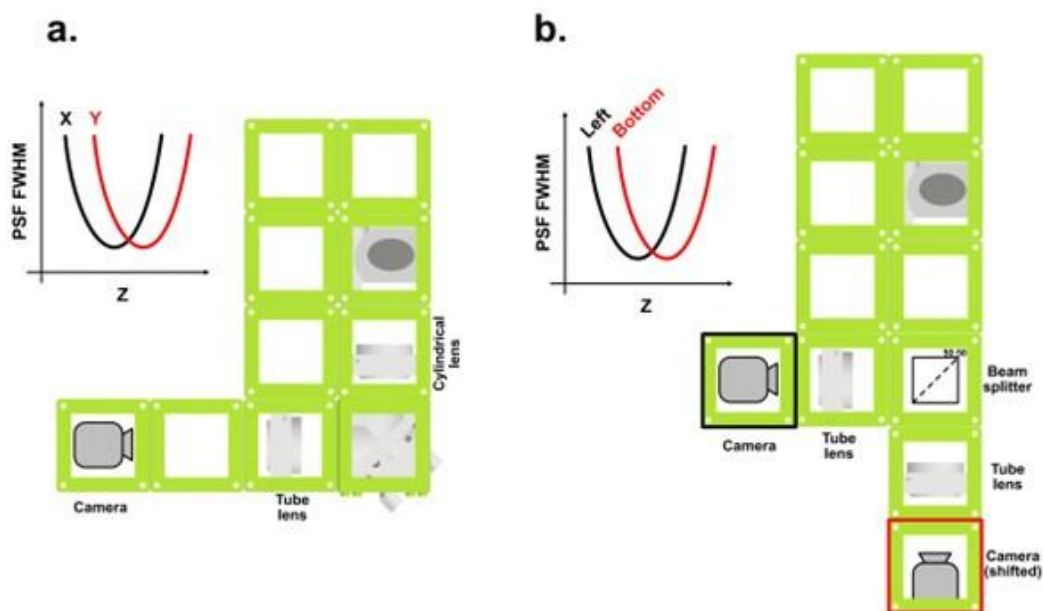
The SPT experiments were performed with QDs to make use of their photostability and brightness. The dSTORM experiment suggests that imaging single AF647 molecules with sufficient SNR is possible. The usage of AF647 for SPT is possible but would reduce the accuracy of the localization and shorten the acquired track-lengths, due to the blinking properties of AF647 when illuminated. The tracking experiment with the QDs thus was a proof of concept that the SPT experiments are possible and allow to reproduce the diffusion coefficients published in common literature. The usage of alternative tags can be an interesting extension of the capabilities that users can perform for their applications.

Figure 6: The authors demonstrate dSTORM of microtubules. It would enhance the paper to also demonstrate 3D imaging (e.g., via cylindrical lens).

The usage of a cylindrical lens for 3D imaging was not performed yet. The implementation would not be difficult, given the high modularity of the setup in general. The calibration of the PSF shape with astigmatism might however be challenging as the vertical scanning of the Z-stage lacks reliability in its current build. Methods such as biplane imaging might also be difficult to implement, as the halved number of photons in each channel leads to losses in the accuracy of localization. As a future improvement of the setup, the option of providing 3D information with single molecule accuracy is definitely desirable and will be tried out. In the following figure, two concepts for introducing 3D imaging capabilities in the detection layer of the microscope are presented.

Author response image 4.

3D concept Figure: Two possible setup modifications to provide axial information when imaging single molecules. a. A cylindrical lens can be placed to induce an asymmetry between the PSF FWHM in x and in y. Every Z position can be identified by two distinct PSF FWHM values in X and Y. b. By splitting the beam in two and defocusing one path, every PSF will have a specific set of values for its FWHM on the two detectors.

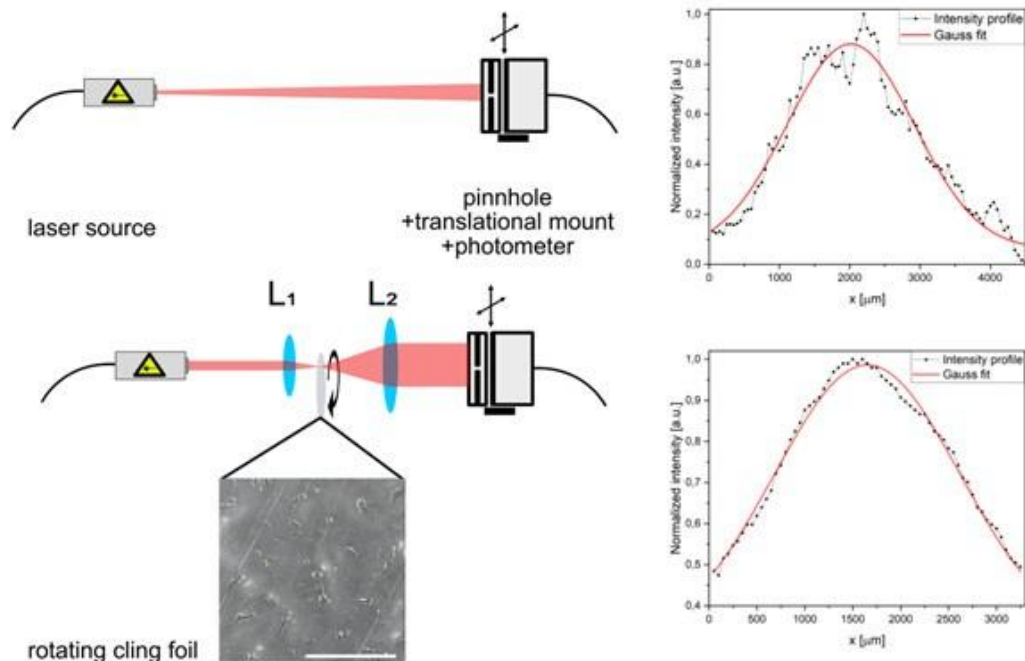


Imaging modalities section: Regarding the use of cling film to diffuse; can the authors comment on the continual use of this approach, including its degradation over time?

The cling foil was only used as a diffuser for broadening the laser profile. A detailed analysis of the constitution of the foil was not done, as no visible changes could be seen on the illumination pattern and the foil itself. The piece of cling foil is attached to a rotor. Detaching of the cling foil or vibrations originating from the rotor need to be minimized. By keeping the rotation speed to a necessary minimum and attaching the cling foil correctly to the rotor, a usable solution can be created. The low price of the cling foil provides the possibility to exchange the foil on a regular basis, allowing to keep the foil under optimal conditions.

Author response image 5.

Profile Figure: By moving a combination of pinhole and photometer to scan through the laser profile with a translational mount, the shape of the laser beam can be estimated. The cling foil plays the same role as a diffuser in other setups.



Reviewer #2 (Recommendations for The Authors):

lines

20, add "," after parts

110, rotating cling foil?

112/116, "custom 3D printed" I thought they were injection molded, please finalize

113, "puzzle pieces" rephrase and they are also barely visible

119, not clear that the stage is a manual stage that was turned into a motorised one by adding belts

123-126, detail for SI,

132, replace Arduino-coded with Arduino-based

143, add reference to Napari

146, (black) cardboard seems to be a cheaper and quicker alternative

153, dichroic

151-155, reads more like a blog post than a paper (maybe add a section on trouble shooting)

156, antibody?

167/189, moderate, please be specific

194, layer of foam material, specify

221, add description/reference to GPI. What is that? why is it relevant?

226: add one sentence description of MMS

318, add "," after students

332-334, as mentioned earlier, not clear, you bought a manual stage and connected belts, correct?

376-377, might be difficult to understand for the layman

391, what laser was used?

Figure 1, poor contrast between components, components visible should be named as much as possible, maybe provide the base layer in a different shade. To me, the red and blue labels look like fluorophores.

Figure 1. looks like d is the excitation layer and not e, please fix.

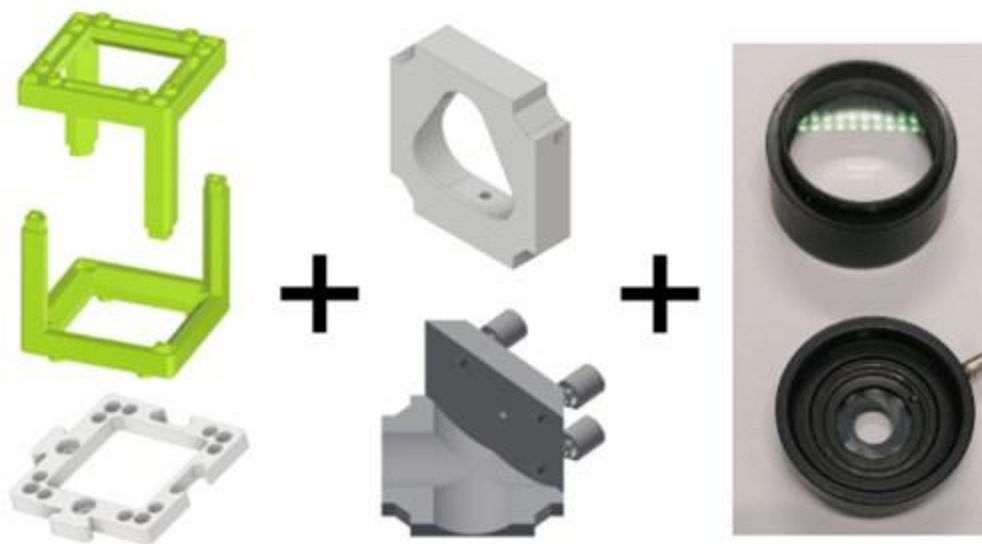
Figure 2, caption a-c, figure 1-d!, btw, why is the drift so anti-correlated?

Figure 6 (line 259) nanometer I guess, not micrometer

We now incorporated all the above-mentioned changes in the manuscript. Furthermore we added the supplementary Figures as below.

Author response image 6.

Basic concept of the UC2 setup: Left: Cubes (green) are connected to one another via puzzle pieces (white). Middle: 3D printed mounts have been designed to adapt various optics (right) to the cube framework. Combined usage of cubes and design of various mounts allows to interface various optics for the assembly.



Author response image 7.

Building the UC2 widefield microscope: a. Photograph of the complete setup. b. All pieces necessary to build the setup. A list of the components can be found in the bill of materials. c. Bottom emission layer of the microscope before assembly. d. Emission layer after assembly. Connection between cubes is doubled by using a layer of puzzles on the top and the bottom of the emission layer. e. CAD schematic of the emission layer and the positioning of the optics. f.

Middle excitation layer of the microscope before assembly. Beam magnifier and homogenizer have been left out for clarity. g. Excitation layer after assembly is also covered by a puzzle layer. h. CAD schematic of the excitation layer and the positioning of the optics. i. Z-stage photograph and corresponding CAD file. Motor of the stage is embedded within the bottom cube. j. A layer of empty cubes supports the microscope stage. k. At this stage of the assembly, the objective is screwed into the objective holder. l. Finally, the stage is wired to the electronics and can then be mounted on top of the microscope (see a.).



Author response image 8.

Measurements performed on the UC2 setup with lower budget objectives. The imaged sample is HeLa cells, stably transfected to express CLC-GFP, then labelled with AF647 through immunostaining. The setup has been kept identical except for the objectives. Scale bar respectively represents 30 μm .

